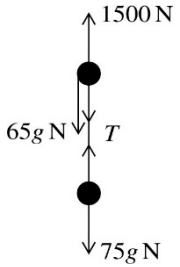
Question			Answer	Marks	AOs	Guidance	
10	(a)		The size/shape of the women are not taken into account	E1 [1]	3.3	Allow women are modelled as point masses	Allow model only involves their mass/weight
10	(b)		The mass (weight) of the rope is negligible [compared to that of the women]	E1 [1]	3.3	Allow either mass or weight used. Allow "no weight" oe	
10	(c)			B1 B1 [2]	3.3 3.3	Draw and label the correct weights Draw and label the given tension and the tension in the second rope. No extra forces	
10	(d)		Total mass 140 kg $1500-65g-75g=140a$	M1 A1 [2]	3.3 1.1	Attempt to form N2L equation with (65+75)*a*. Condone one missing force All correct	
10	(e)		$a=\frac{32}{35}=0.914\ \text{m s}^{-2}$	B1 [1]	1.1	cao Allow from the solution of two separate equations	(See below for correct separate equations)
10	(f)		N2L for rescued woman: $T-75g=75\times\frac{32}{35}$ $T=\frac{5625}{7}=804\ \text{N}$	M1 A1 A1	3.1b 1.1 1.1	Attempt to form an equation for the rescued woman (mass 75 kg) Correct equation FT their *a* Correct value for *T*. Allow for 800 N from a correct equation	Equation must include *T*
			Alternative solution N2L for rescue worker: $1500-65g-T=65\times\frac{32}{35}$ $T=\frac{5625}{7}=804\ \text{N}$	M1 A1 A1		Attempt to form an equation for the rescue worker (mass 65 kg) Correct equation FT their *a* Correct value for *T*. Allow for 800 N from a correct equation	Equation must include *T*
				[3]			